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CROSS-RATE CALCULATIONS

Learning Outcome

- calculate and interpret currency cross-rates

Global currencies are bought, sold, and exchanged in the foreign exchange (FX) market. In this decentralized market, participants trade currencies utilizing exchange rates, which typically reflect an efficient market. This section will cover the use of cross exchange rate relationships (cross-rates) to calculate exchange rates between two currencies using a third currency. It also will introduce calculations used in the FX market to trade currencies.

Given two exchange rates involving three currencies, it is possible to back out the cross-rate. For example, as we have seen in a prior lesson, the FX market convention is to quote the exchange rate between the US dollar and the euro as euro–dollar (USD/EUR). The FX market also quotes the exchange rate between the Canadian dollar and US dollar as dollar–Canada (CAD/USD). Given these two exchange rates, it is possible to back out the cross-rate between the euro and the Canadian dollar, which according to market convention is quoted as euro–Canada (CAD/EUR). This calculation is shown as follows:

$$\frac{\text{CAD}}{\text{USD}} \times \frac{\text{USD}}{\text{EUR}} = \frac{\text{CAD}}{\cancel{\text{USD}}} \times \frac{\cancel{\text{USD}}}{\text{EUR}} = \frac{\text{CAD}}{\text{EUR}}.$$

Hence, to get a euro–Canada (CAD/EUR) quote, we must multiply the dollar–Canada (CAD/USD) quote by the euro–dollar (USD/EUR) quote. For example, assume the exchange rate for dollar–Canada is 1.3020 and the exchange rate for euro–dollar is 1.1701. Using these spot exchange rates, the euro–Canada cross-rate equals:

1.3020 × 1.1701 = 1.5235 CAD per EUR.

The professional FX market does not use the convention of direct or indirect quotes because these conventions depend on one's location to determine the domestic versus foreign currencies. Instead, the market uses rate quotes on defined conventional currency pairs. Sometimes, to get a cross-rate using several currency quotes, it is necessary to invert a quote to get an intermediary currency that can be canceled out in the equation to obtain the cross-rate. For example, to get a Canada–

the dollar–Canada rate (CAD/USD) be inverted to a Canada–dollar (USD/CAD) quote for the calculations to work, as follows:

$$\left(\frac{\text{CAD}}{\text{USD}}\right)^{-1} \times \frac{\text{JPY}}{\text{USD}} = \frac{\text{USD}}{\text{CAD}} \times \frac{\text{JPY}}{\text{USD}} = \frac{\cancel{\text{USD}}}{\text{CAD}} \times \frac{\text{JPY}}{\cancel{\text{USD}}} = \frac{\text{JPY}}{\text{CAD}}$$

Hence, to get a Canada–yen (JPY/CAD) quote, we must first invert the dollar–Canada (CAD/USD) quote before multiplying by the dollar–yen (JPY/USD) quote. Market quotes for most currencies are quoted to four decimal places; however, the Japanese yen exchange rate is quoted to two decimal places. For example, assume that we have spot exchange rates of 1.3020 for dollar–Canada (CAD/USD) and 111.94 for dollar–yen (JPY/USD). The dollar–Canada rate of 1.3020 inverts to 0.7680; multiplying this value by the dollar–yen quote of 111.94 gives the following Canada–yen quote:

0.7680 × 111.94 = 85.97 JPY per CAD.

Market participants asking for a quote in a cross-rate currency pair typically will not need to do this calculation themselves: Either the dealer or the electronic trading platform will provide a quote in the specified currency pair. (For example, a client asking for a quote in Canada–yen will receive that quote from the dealer; he will not be given separate dollar–Canada and dollar–yen quotes to do the calculation.) Dealers providing the quotes often have to do this calculation themselves if only because the dollar–Canada and dollar–yen currency pairs often trade on different trading desks and involve different traders. Electronic dealing machines used in both the interbank market and bank-to-client markets often provide this mathematical operation to calculate cross-rates automatically.

Because market participants can receive both a cross-rate quote (e.g., Canada–yen) as well as the component underlying exchange rate quotes (e.g., dollar–Canada and dollar–yen), these cross-rate quotes must be consistent with the previous equation; otherwise, the market will arbitrage the mispricing. Extending our example, we calculate a Canada–yen (JPY/CAD) rate of 85.97 based on underlying dollar–Canada (CAD/USD) and dollar–yen (JPY/USD) rates of 1.3020 and 111.94, respectively. Now suppose that at the same time a misguided dealer quotes a Canada–yen rate of 86.20. This is a different price in Canada–yen for an identical service—that is, converting yen into Canadian dollars. Hence, any trader could buy CAD1 at the lower price of JPY85.97and then turn around and sell CAD1 at JPY86.20 (recall our earlier discussion of how price and base currencies are defined). The riskless arbitrage profit is JPY0.23 per CAD1. The arbitrage—called *triangular arbitrage* (we use

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occur because both human traders and automatic trading algorithms are constantly on alert for any pricing inefficiencies. In practice, and for the purposes of this lesson, we can consider cross-rates as being consistent with their underlying exchange rate quotes and can assume that given any two exchange rates involving three currencies, we can back out the third cross-rate.

EXAMPLE 1

Cross-Rates and Percentage Changes

A research report produced by a dealer includes the following spot rate quotes:

Currency	Spot Rate	Expected Spot Rate in One Year
USD/EUR	1.1701	1.1619
CHF/USD	0.9900	0.9866
USD/GBP	1.3118	1.3066

1. The spot CHF/EUR cross-rate is *closest* to:

A. 0.8461.
B. 0.8546.
C. 1.1584.

Hide Solution

Solution:

C is correct:

$$\frac{\text{CHF}}{\text{EUR}} = \frac{\text{USD}}{\text{EUR}} \times \frac{\text{CHF}}{\text{USD}} = 1.1701 \times 0.9900 = 1.1584$$

2. The spot GBP/EUR cross-rate is *closest* to:

A. 0.8920.
B. 1.1211.
C. 1.4653.

Hide Solution

Solution:

A is correct:

$$\frac{\text{GBP}}{\text{EUR}} = \frac{\text{USD}}{\text{EUR}} \times \left(\frac{\text{USD}}{\text{GBP}}\right)^{-1} = \frac{\text{USD}}{\text{EUR}} \times \frac{\text{GBP}}{\text{USD}} = \frac{1.170}{1.311}$$

3. Based on the research report, the euro is expected to appreciate by how much against the US dollar over the

C. +1.0 percent

Hide Solution

Solution:

A is correct. The euro is the base currency in the USD/EUR quote, and the expected decrease in the USD/EUR rate indicates that the euro is depreciating. In one year, it will cost less, in US dollars, to buy one euro. Mathematically:

$$\frac{1.1619}{1.1701} - 1 = -0.7\%$$

4. Based on the research report, how much is the US dollar expected to appreciate against the British pound sterling over the next year?

A. +0.6 percent
B. –0.4 percent
C. +0.4 percent

Hide Solution

Solution:

C is correct. The British pound is the base currency in the USD/GBP quote, and the expected decrease in the USD/GBP rate means that the British pound is expected to depreciate against the US dollar. Or equivalently, the US dollar is expected to appreciate against the British pound. Mathematically:

$$\left(\frac{1.3066}{1.3118}\right)^{-1} - 1 = \frac{1.3118}{1.3066} - 1 = +0.4\%$$

5. Over the next year, the Swiss franc is expected to:

A. depreciate against the British pound.
B. depreciate against the euro.
C. appreciate against the British pound, euro, and US dollar.

Hide Solution

Solution:

C is correct: Because the question does not require calculating the magnitude of the appreciation or depreciation, we can use the Swiss franc as either the price currency or the base currency. In this case, it is easier to use the Swiss franc as the price currency. CHF/USD is expected to decline from 0.9900 to 0.9866, so the Swiss franc is expected to

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(= 0.9866 × 1.1619), so the Swiss franc is expected to appreciate against the euro. CHF/GBP is currently 1.2987 (= 0.9900 × 1.3118) and is expected to be 1.2891 (= 0.9866 × 1.3066), so the Swiss franc is also expected to appreciate against the British pound.

Alternatively, we can derive this answer intuitively. According to the research report, the CHF/USD rate is expected to decline: That is, the US dollar is expected to depreciate against the Swiss franc, or alternatively, the Swiss franc is expected to appreciate against the US dollar. The USD/EUR and USD/GBP rates are also decreasing, meaning that the euro and British pound are expected to depreciate against the US dollar, or alternatively, the US dollar is expected to appreciate against the euro and British pound. If the Swiss franc is expected to appreciate against the US dollar and the US dollar is expected to appreciate against both the euro and British pound, it follows that the Swiss franc is expected to appreciate against both the euro and British pound.

6. Based on the research report, which of the following lists the three currencies from strongest to weakest over the next year?

A. US dollar, British pound, euro
B. US dollar, euro, British pound
C. Euro, US dollar, British pound

Hide Solution

Solution:

A is correct. USD/EUR is expected to decline from 1.1701 to 1.1619, while USD/GBP is expected to decline from 1.3118 to 1.3066. So, the US dollar is expected to be stronger than both the euro and British pound. GBP/EUR is currently 0.8920 [= $(1.3118)^{-1}$ × 1.1701] and is expected to be 0.8893 [= $(1.3066)^{-1}$ × 1.1619], so the British pound is expected to be stronger than the euro.

7. Based on the research report, which of the following lists the three currencies in order of appreciating the most to appreciating the least (in percentage terms) against the US dollar over the next year?

A. British pound, Swiss franc, euro

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Solution:

B is correct. The USD/EUR rate depreciates by –0.7 percent (= [1.1619/1.1701] – 1), which is the depreciation of the base currency euro against the US dollar. The USD/GBP rate declines –0.4 percent (= [1.3066/1.3118] – 1), which is the depreciation of the British pound against the US dollar. Inverting the CHF/USD rate to a USD/CHF convention shows that the base currency Swiss franc appreciates by +0.35 percent against the US dollar (= [1.0136/1.0101] – 1).

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AP

Q6 - Consistent mistake in quoting exchange rate as the quotations in the text deviates from the standard conventions. Question 6 has the wrong exchange rates given - wrong conventions used etcFor instance, the EUR and the GBP are both stronger currencies than USD and therefore, the standard exchange rates quoted with USD as the quoted currency and the EUR as the base currency would read like:EUR/USD - For today, for instance, this is 1.17 which means 1.17 USDs exchage for 1 EURorGBP/USD - For today, for instance, this is 1.36 which means 1.36 USDs exchage for 1 GBPIf we were to swap the base currency with the quoted currency, we would get the exchange rate in fractions which still makes the EUR and GBP stronger. Exactly the opp is the rationale provided in Q6, and there is NO correct answer given in the choices

Tags: error

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Created a month ago by Anoosha Pervaz

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OK

Where exactly is explained a situation where we have 4 currencies which involves multiplying and dividing of different ones (as we have in practice questions after this learning topic)?
For example having USD/EUR, CAD/USD, CHF/EUR. And we need CAD/CHF.
I struggle to find that in learning module, how I am supposed to answer such practice question?

Created 2 months ago by Oleksandr Kyrychuk

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FA

I think there is a mistake at question 6

Created 7 months ago by Francois Abillama

5 replies | Last Activity: 2 months ago

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MO

Q3 is incorrect, it's asking how much the euro will appreciate, however it should be asking how much the dollar will appreciate, as the result shows that the euro is in fact depreciating, not appreciating.

Created 5 months ago by Maxim Ovchinnikov

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